Writing a Paper in R!

Christelinda Laureijs

This lecture is a how-to guide for:

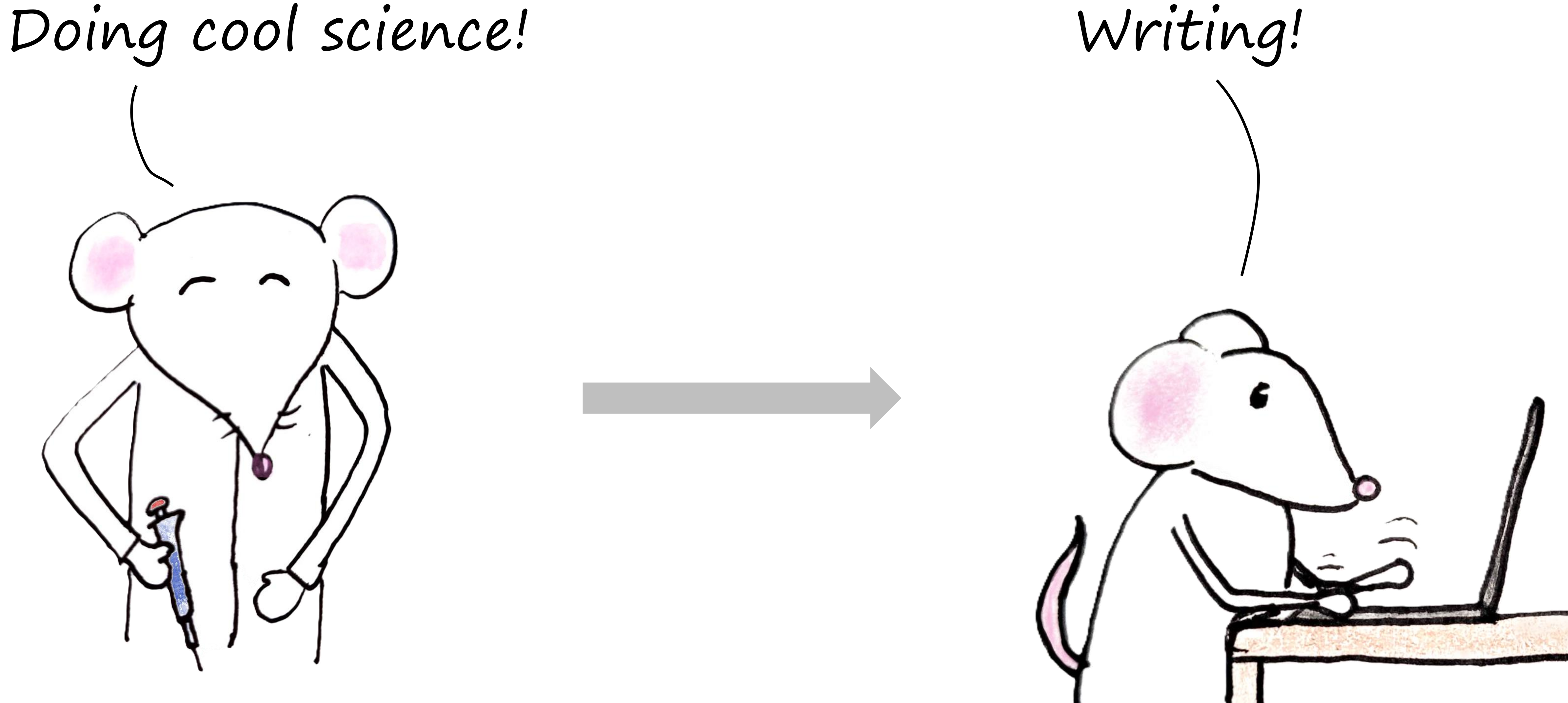
✓ Figures + captions

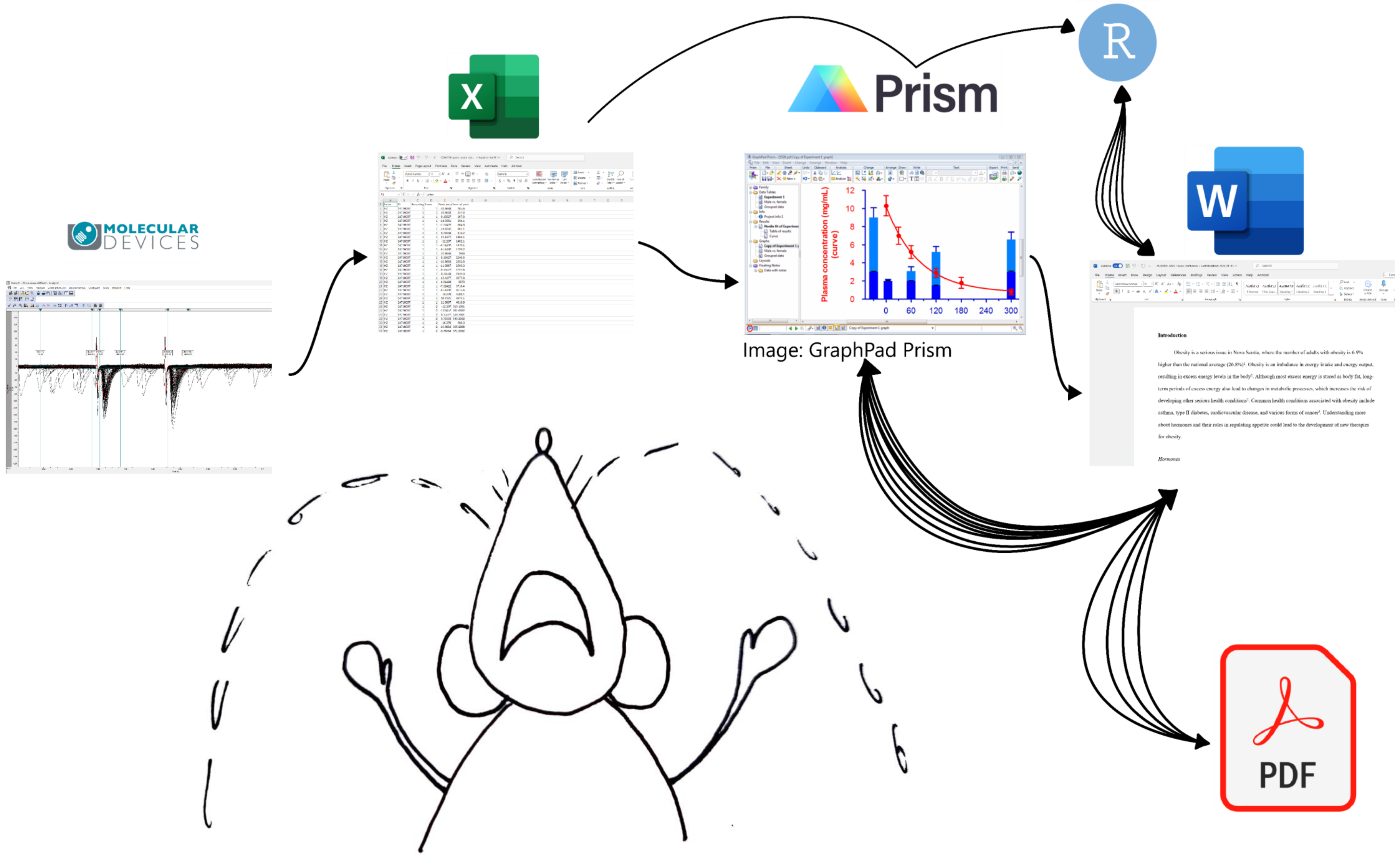
✓ Inline statistics and tables

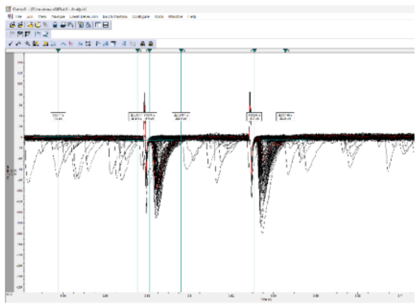
✓ Table of Contents, list of figures …etc.

✓ Basic LaTeX customization

✓ Citations


Doing cool science!
Writing!


MOLECULAR DEVICES
Prism
Plasma concentration (mg/mL) (curve)
Image: GraphPad Prism
Introduction
PDF

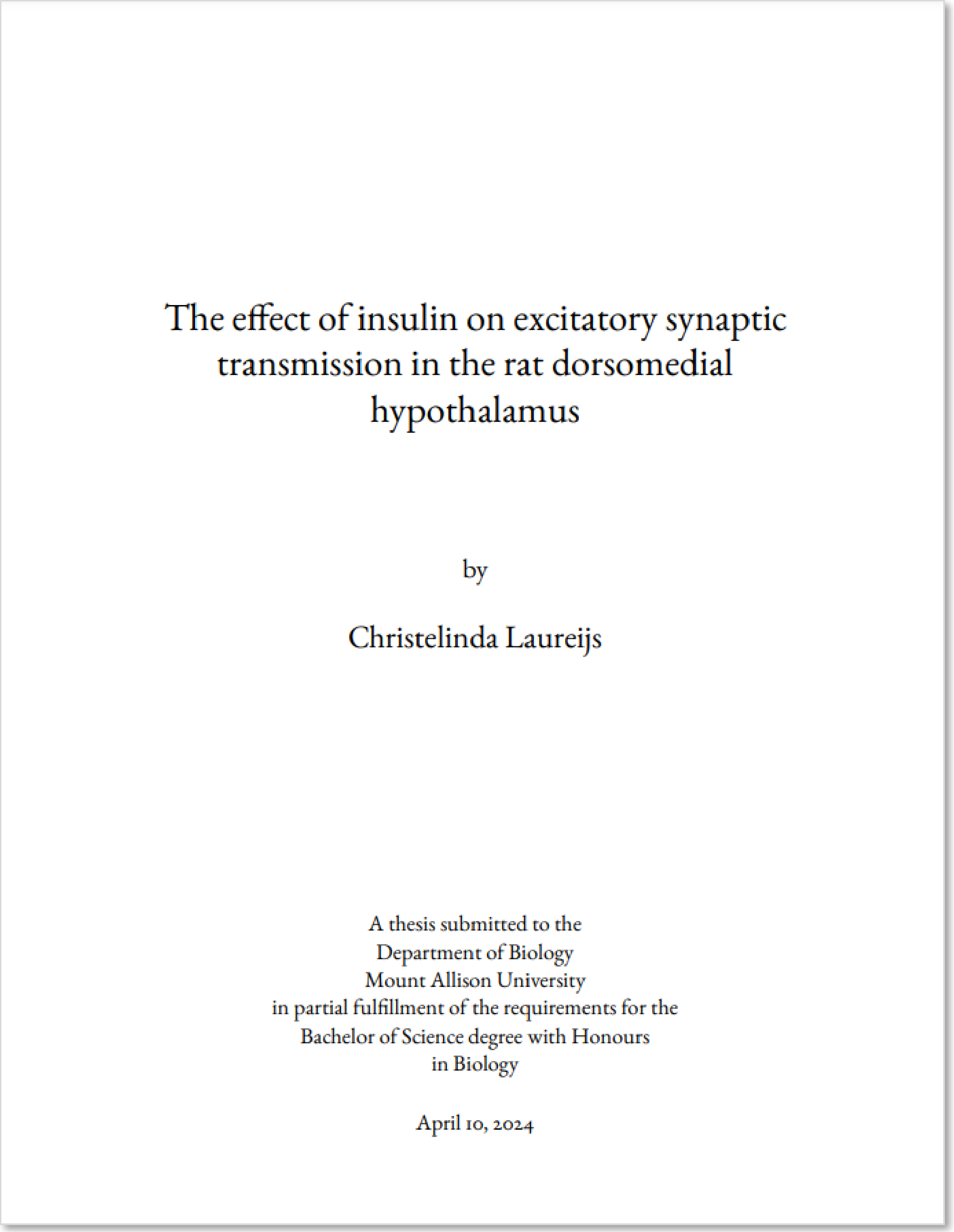


- ✓ More efficient
- ✓ Fully reproducible
- ✓ NO MORE manual formatting!!
- ✓ Plaintext files – no proprietary software, small size
- ✓ Track with Git

CASE STUDY

My honours thesis

https://github.com/christelinda-laureijs/honours-thesis



The effect of insulin on excitatory synaptic transmission in the rat dorsomedial hypothalamus

by

Christelinda Laureijs

A thesis submitted to the
Department of Biology
Mount Allison University
in partial fulfillment of the requirements for the
Bachelor of Science degree with Honours
in Biology

April 10, 2024

Low maintenance and interactive!

Contents

Auto-numbering

Heading 1 →

Methods

Heading 2 →

Animals

I used young (28-44 days old) male and female Sprague-Dawley rats from Charles River Laboratories (Québec, Canada) for all treatment groups. The rats arrived when they were 22 days old, and they had a one-week acclimation period. The rats lived in groups of 4 with free access to standard rat chow and water. All cages had wood shavings for bedding, toys (wooden blocks and a plastic chew toy) for enrichment and Crawl Balls™ (Bio-Serv) for shelter. The animal rooms were maintained at 21°C and 50% ± 10% humidity, with a 12-hour light/dark cycle. The Animal Care Committee at Mount Allison University approved the project (Protocol #103088).

Heading 2 →

Brain Removal & Slice Preparation

Québec,

Advanced typography,

e.g. contextual alternates

&

Full justification with no rivers

Methods

Animals

I used young (28-44 days old) male and female Sprague-Dawley rats from Charles River Laboratories (Québec, Canada) for all treatment groups. The rats arrived when they were 22 days old, and they had a one-week acclimation period. The rats lived in groups of 4 with free access to standard rat chow and water. All cages had wood shavings for bedding, toys (wooden blocks and a plastic chew toy) for enrichment and Crawl Balls™ (Bio-Serv) for shelter. The animal rooms were maintained at 21°C and 50% ± 10% humidity, with a 12-hour light/dark cycle. The Animal Care Committee at Mount Allison University approved the project (Protocol #103088).

Brain Removal & Slice Preparation

Auto-updating p-values!

nsulin significantly de-

d ($p < 0.001$; Figure 8).

Automated cross-referencing

ne first five minutes of

Auto-updating tables & results

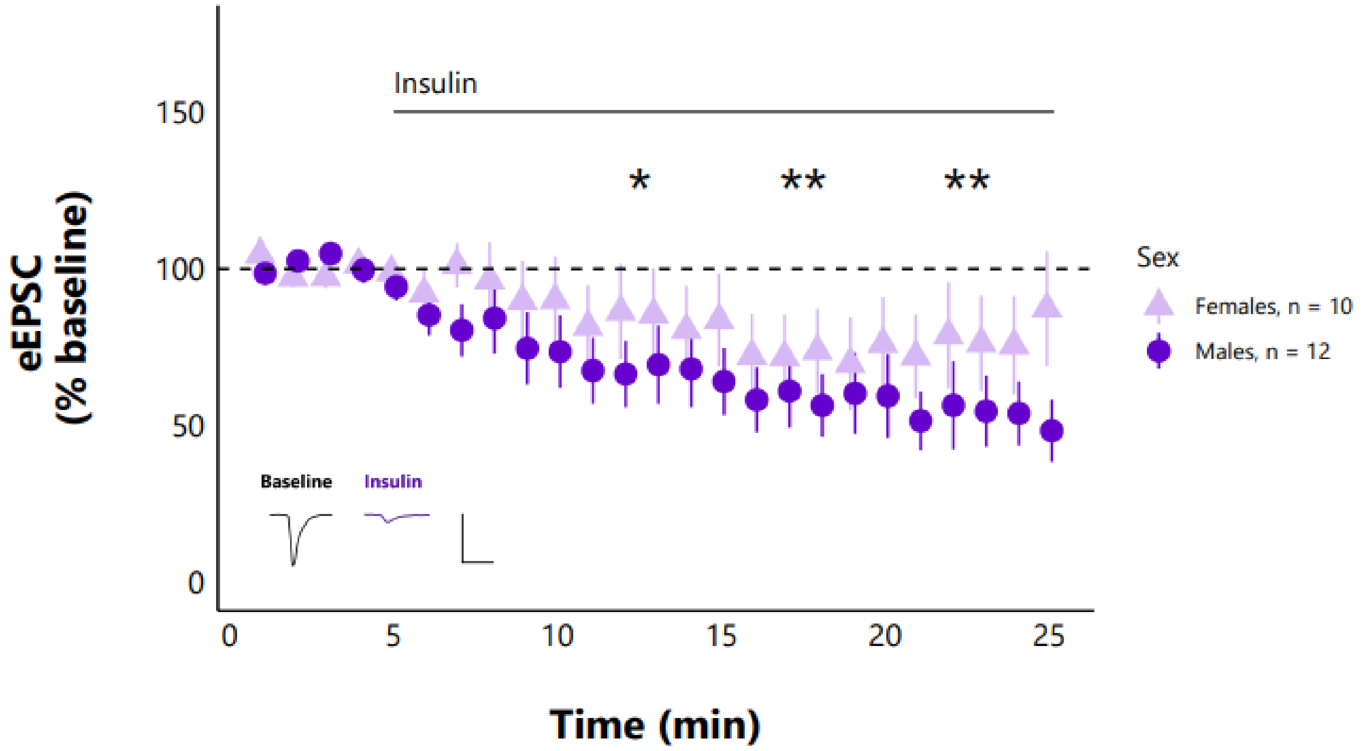
Table 1: A Wilcoxon signed-rank test shows that insulin significantly affects action potential amplitudes, thresholds, and half-widths. Insulin does not significantly affect the latency to fire, after-hyperpolarization amplitude, or the time of afterhyperpolarization.

Parameter	df	Statistic	*p*-value
Peak amplitude	15, 15	120	< 0.001
Threshold	15, 11	63	0.005
Latency to fire	15, 11	31	0.90
Half-width	15, 11	2	0.003
After-hyperpolarization amplitude	15, 11	35	0.90
After-hyperpolarization time	15, 11	17	0.17

In-text statistics

Insulin significantly decreases action potential amplitude relative to baseline amplitudes ($W(15, 15) = 120$, $p = < 0.001$; Figure 7C). In four recordings, there were no amplitudes after insulin exposure, so additional action potential parameters could not be measured. Of the action potentials that did fire after insulin exposure, the threshold decreased significantly ($W(15, 11) = 63$, $p = 0.005$; Figure 7D). Action potential

List of Figures



Caption

Same figure: different caption than in the list of figures!

Figure 8: Insulin significantly decreases excitatory evoked post-synaptic current (eEPSC) amplitude over time in both sexes, with no significant differences between the sexes. I exposed DMH neurons to 500 nM of insulin from 5 minutes and onward. Each point represents the mean eEPSC amplitude (± the standard error) across all cells and *n* represents the number of unique cells. The asterisks indicate a statistically significant decrease in current amplitude relative to the baseline (t-test; * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$). The representative traces consist of eEPSCs from one cell averaged over the baseline period (0 to 5 min) and last interval (20 to 25 min) of the recording, and the scale bar represents 50 pA/20 ms.

Insulin may bind to insulin-like growth factor 1 receptors on

How to write a paper in R

1. Identify paper **sections** and build **outline** in R

   Knit to pdf!

2. Insert **figures** and captions

   Knit to pdf!

3. Run statistical **analyses**

   Knit to pdf!

4. Insert statistical results as **tables** and **inline-text**

   Knit to pdf!

5. Insert **citations**

   Knit to pdf!

6. Modify **figures**

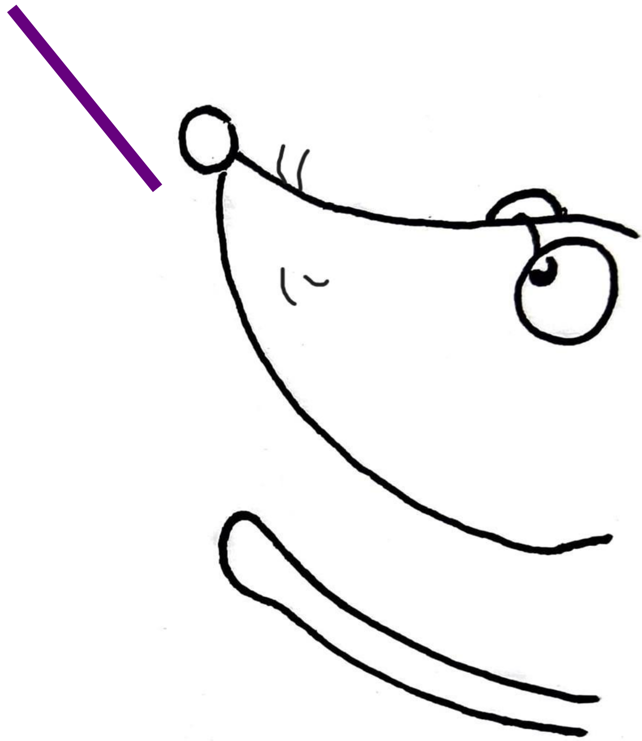
   Knit to pdf!

7. **Customize** font, page layout, margin, etc.

   Knit to pdf!

Setup

Why use R Projects?



masters-thesis.Rproj

README.md

Thesis

Figures

Data

An RProject is your “home” folder

- masters-thesis.Rproj
- README.md
- Thesis
- Figures
- Data

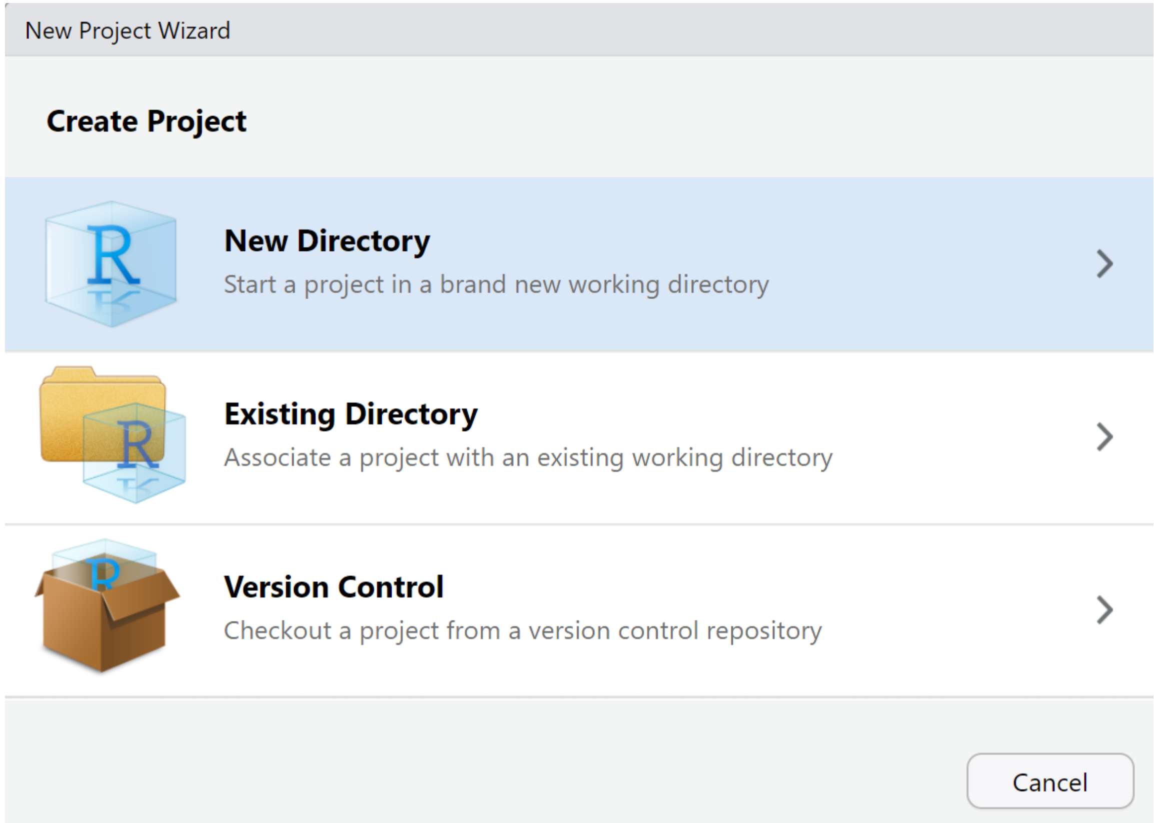
Better than manually setting working directory

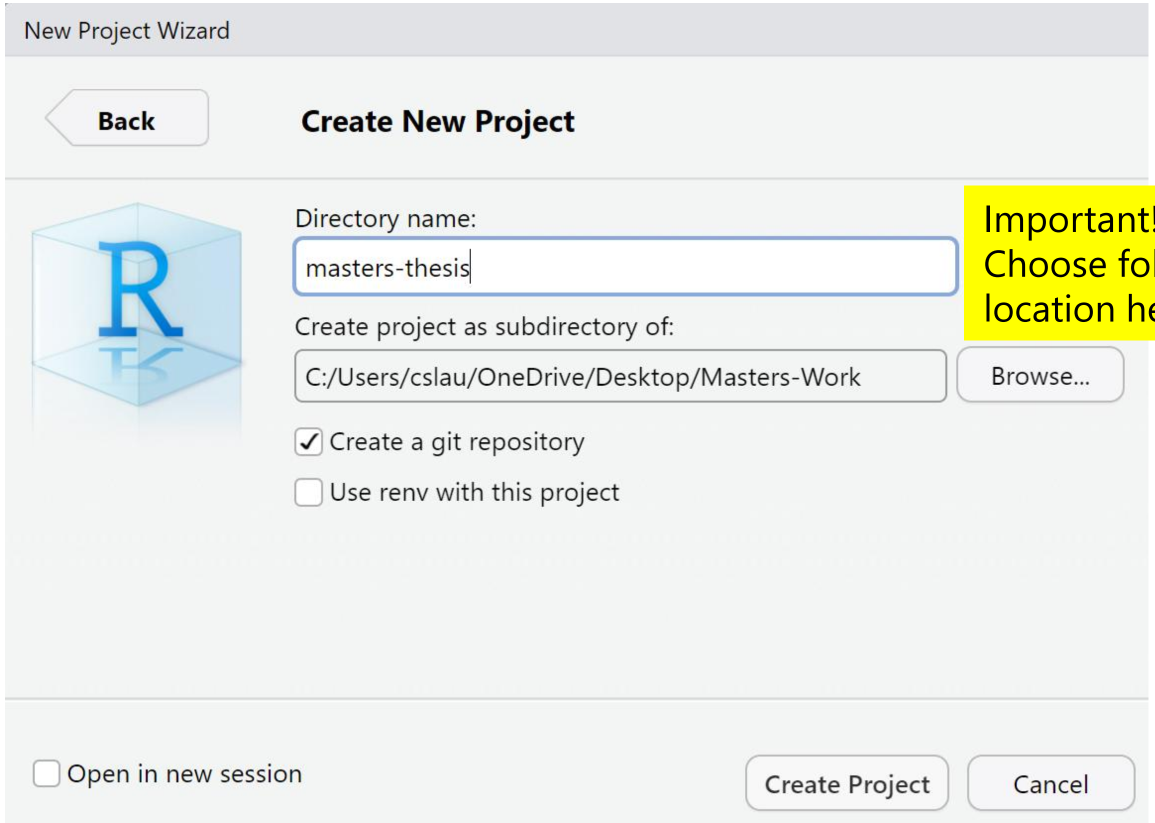
Everything works even on another computer – relative paths

Before: `“C:/Users/Christelinda/Masters/Thesis/Data/Raw-data.csv”`

After: `“Data/Raw-data.csv”`

1. Open an R Project (existing or start a new one)



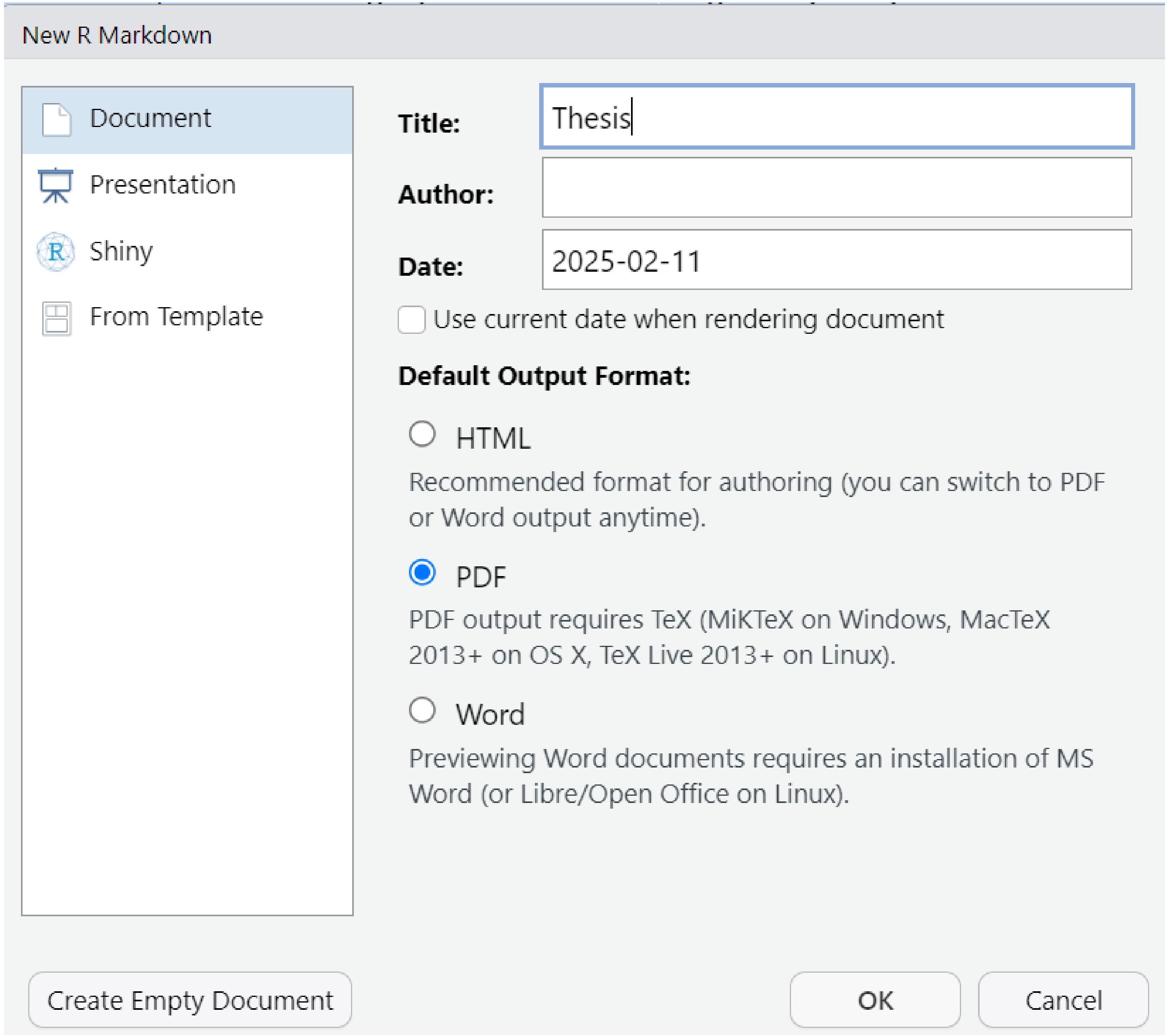


Important! Choose folder location here!

2. File → New file → RMarkdown

a. PDF output

b. "Save as" in the "Thesis" folder



3. Knit to PDF

Ugly, but it works!

Untitled

2025-02-11

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00  
```

Including Plots

You can also embed plots, for example:

4. Delete all filler content, fill with your headers

#* = *Level 1

##* = *Level 2

```
---
title: "Thesis"
output: pdf_document
---

# Introduction

# Methods

## Animals

## Reagents

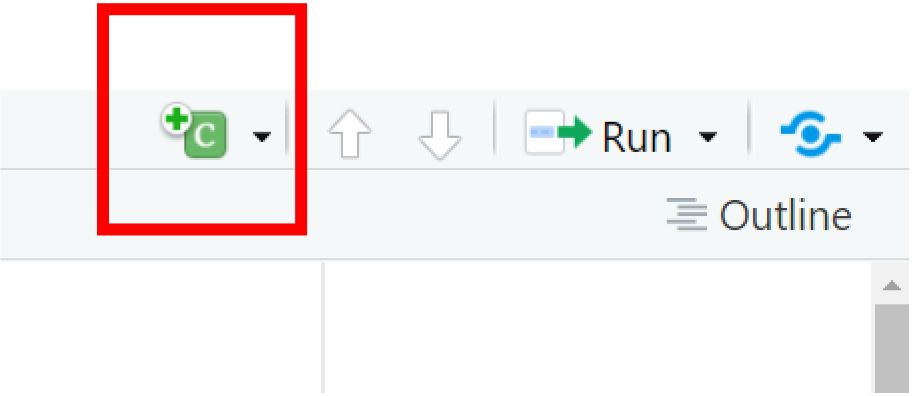
# Results

# Discussion


```

5a. Insert a new code chunk

`Ctrl + Alt + I`



5a. Name it (e.g. "setup") and add echo = FALSE to hide it

```
---
title: "Thesis"
output: pdf_document
---

```{r setup, echo = FALSE}

```
```

5c. Create a set-up chunk

```
```{r setup, echo = FALSE}
knitr::opts_chunk$set(
  dev = c("cairo_pdf"),
  dpi = 600,
  fig.align = "center",
  out.width = "100%",
  comment = NA,
  echo = FALSE,
  message = FALSE,
  warning = FALSE
)
```
```

Searchable text in figures (e.g. axis titles) (*Mac: may need "png"*)

High-quality figure resolution

Set default width of the figures (% total page content width)

Remove ## in front of chunk output

Hide warnings and code chunks in output PDF

Figures

6. Load libraries in a new chunk

```
library(ggplot2)
library(dplyr)
library(here)
```

Make plots

Manipulate data

Get files in subfolders with relative paths

“Data/Raw-data.csv”

In this tutorial, you will also need these:

```
library(broom)
library(stringr)
library(knitr)
```

Get clean ANOVA output in table format

Manipulate text

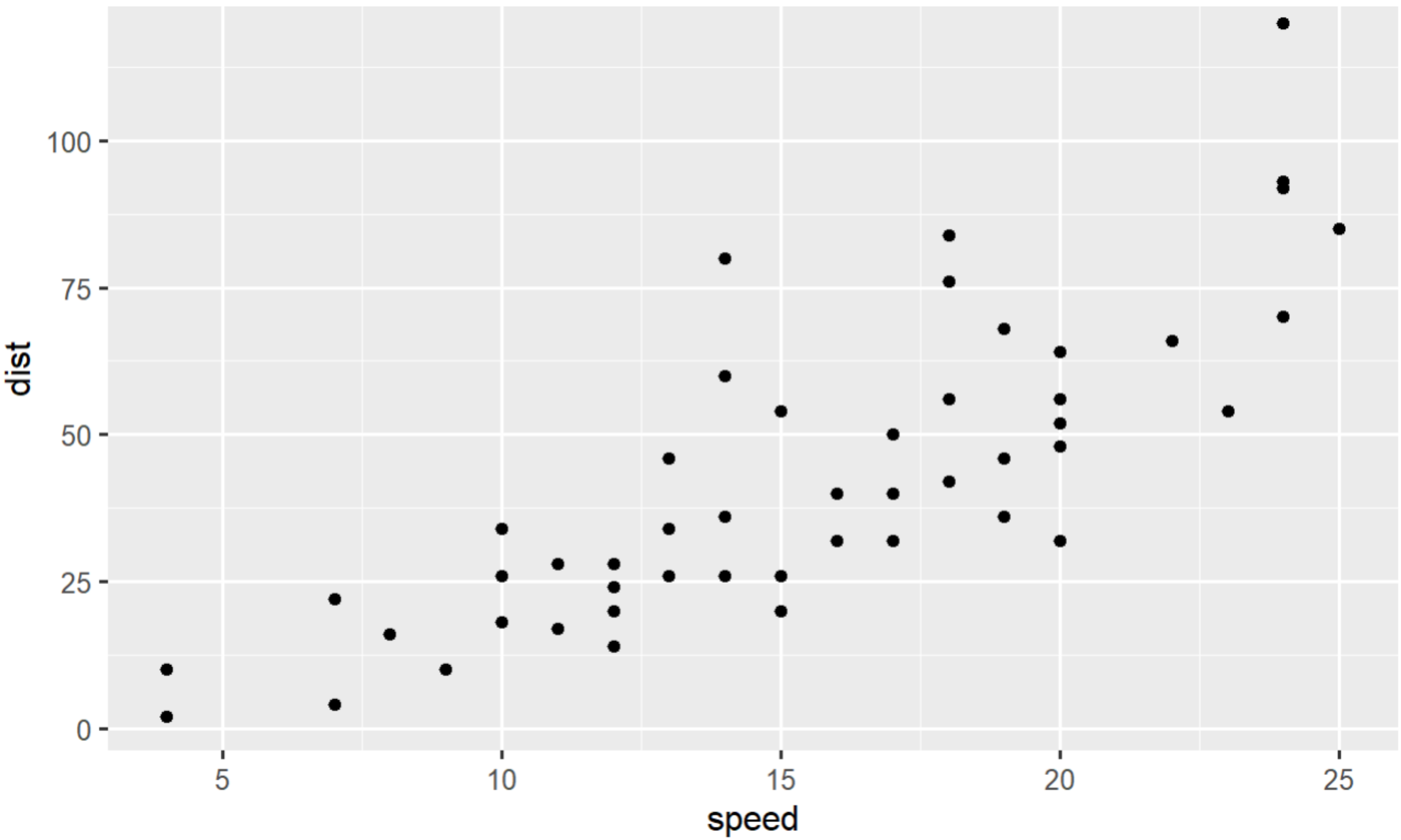
Tools for knitting to PDF

7. Create a simple plot

```
cars %>%
  ggplot(aes(x = speed, y = dist)) +
  geom_point()
```

speed	dist
4	2
4	10
7	4
7	22
8	16
9	10
10	18
10	26

7b. Marvel at its beauty



8. Make plot look better

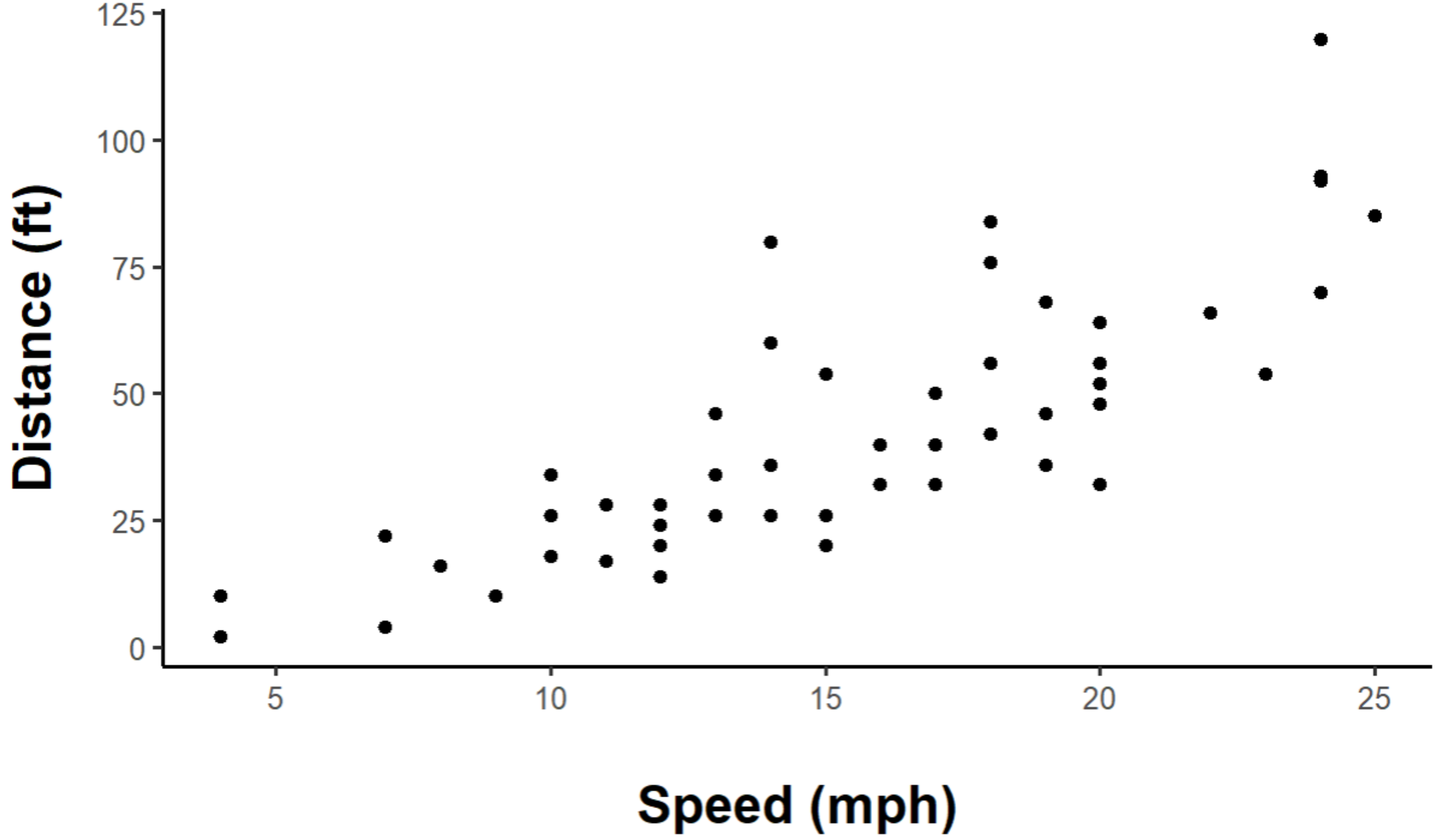
```
cars %>%
  ggplot(aes(x = speed, y = dist)) +
  geom_point() +
  labs(x = "Speed (mph)", y = "Distance (ft)") +
  theme_classic() +
  theme(text = element_text(color = "black"),
        axis.title = element_text(size = 15, face = "bold"),
        axis.title.x = element_text(margin = margin(t = 20)),
        axis.title.y = element_text(margin = margin(r = 10)),
        plot.margin = margin(20, 20, 20, 20))
```

Axis titles

No grid

Better margins

8. Better!



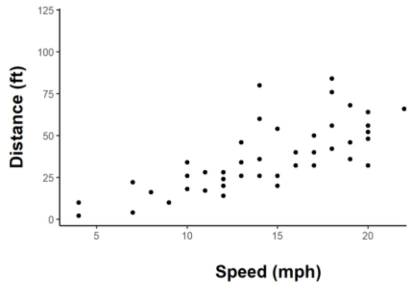
8b. Set default theme for all ggplots

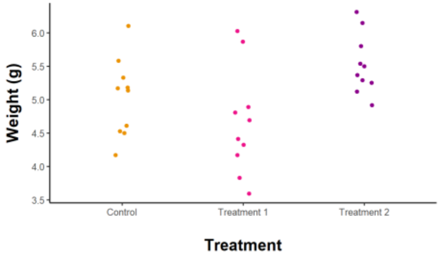
Use theme_set() in a chunk near the top of the document

```
theme_set(theme_classic() +
  theme(text = element_text(color = "black"),
        axis.title = element_text(size = 15, face = "bold"),
        axis.title.x = element_text(margin = margin(t = 20)),
        axis.title.y = element_text(margin = margin(r = 10)),
        plot.margin = margin(20, 20, 20, 20)))
```

Reduces repetitive theme code in all plots

```
cars %>%
  ggplot(aes(x = speed, y = dist)) +
  geom_point() +
  labs(x = "Speed (mph)", y = "Distance (ft)")
```





9. Add caption **(fig.cap)** and short caption **(fig.scap)**

```
```{r cars-plot, fig.cap = "Faster cars take a longer distance to stop.", fig.scap
= "Stopping distance of cars"}
```

10. Add a **\tableofcontents** and **\listoffigures**

```
\tableofcontents

\listoffigures


# Introduction
```
```

11. Knit a PDF! Do NOT worry about formatting, fonts or spacing yet!

Contents

List of Figures

fig.scap

12. Cross-reference image with **\ref{}**

```
```{r cars-plot, fig.cap = "Faster cars take a longer distance to stop.
\\label{cars-plot}", fig.scap = "Stopping distance of cars"}
```

Must put **\\label{}** INSIDE the fig.cap quotations! No spaces!

```
faster cars took longer distances to stop (Figure \ref{cars-plot}).
```

Knit

faster cars took longer distances to stop (Figure 1).

Clickable!
```

12. Cross-reference image with **\ref{}**

```
```{r cars-plot, fig.cap = "Faster cars take a longer distance to stop.
\\label{cars-plot}", fig.scap = "Stopping distance of cars"}
```
```

13. Change other parameters in the chunk options

```
```{r cars-plot}
#| fig.cap = "Faster cars take a longer distance to stop. \\label{cars-plot}",
#| fig.scap = "Stopping distance of cars"
#| out.width = "75%"
```

Use **#|** to extend chunk options into a new line

\\textit{} = italics

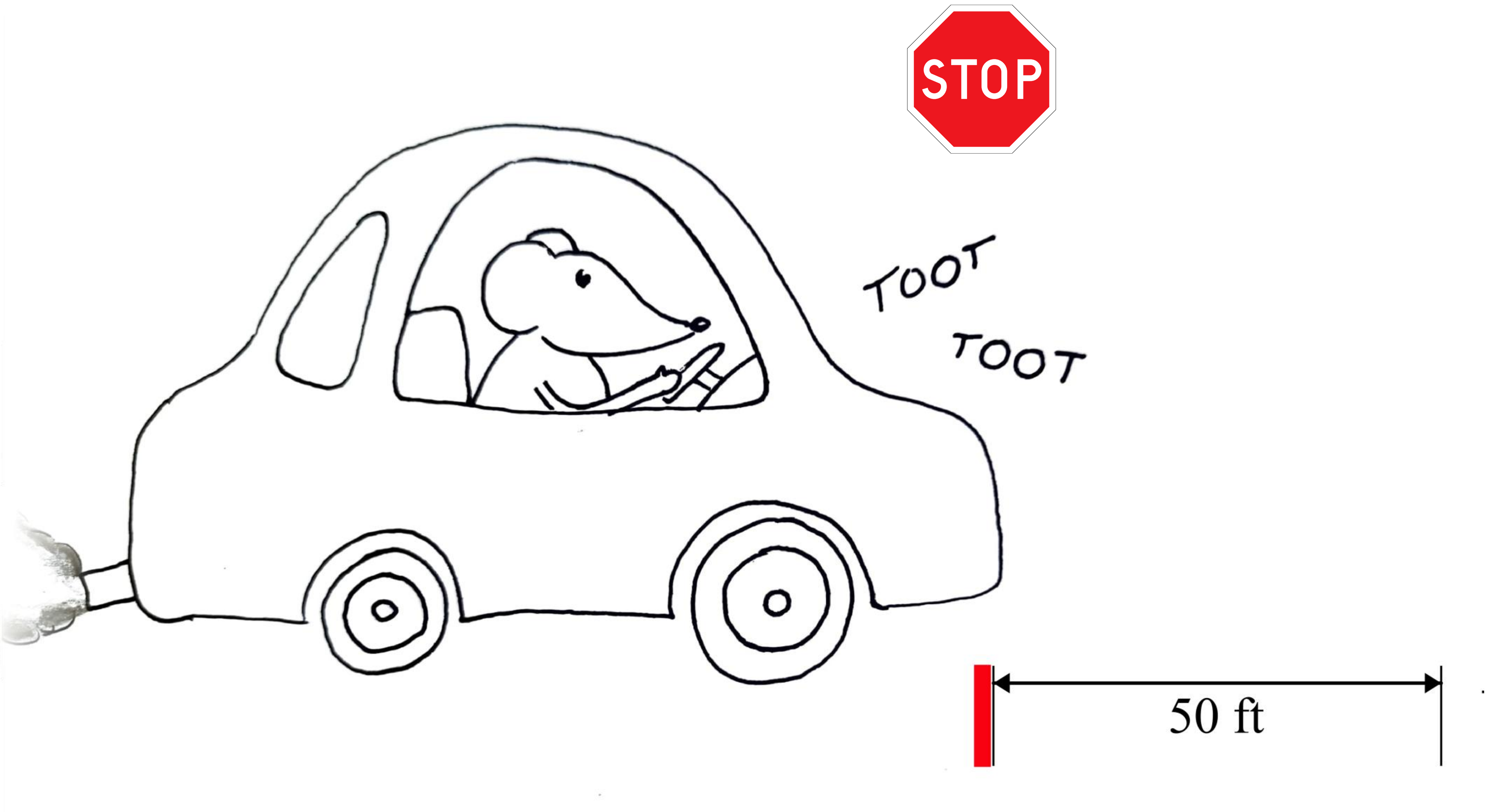
Use \ to cancel special characters like “ or ‘

```
fig.cap = "Faster cars take a \\textit{longer} distance to \"stop\".
```

Figure 1: Faster cars take a *longer* distance to "stop".
```

Methods schematic

* Not to scale



14. Insert images with **knitr::include_graphics()**

```
knitr::include_graphics(here("Figures/Rat-car-schematic.png"))
```

here() selects file in the "Figures" folder

Updated figures numbers

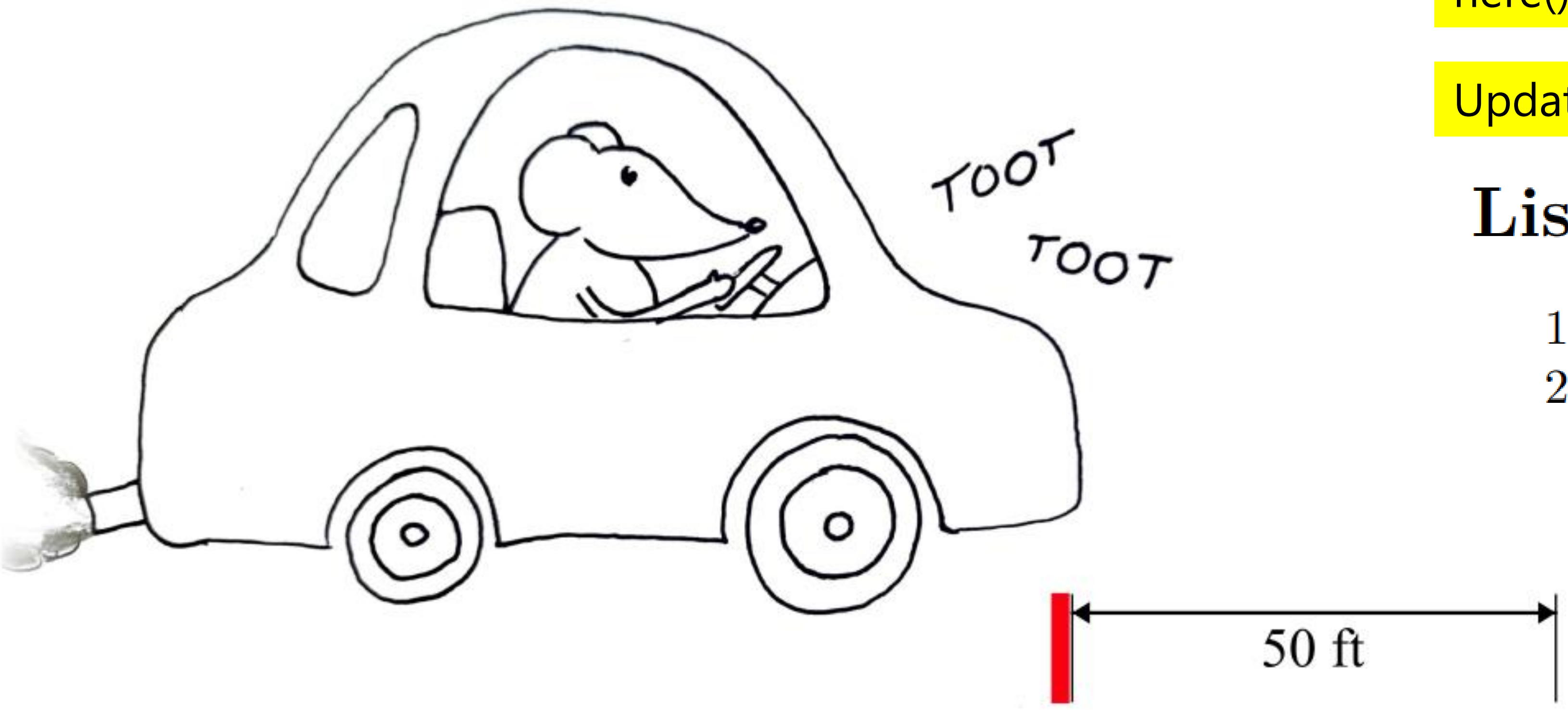


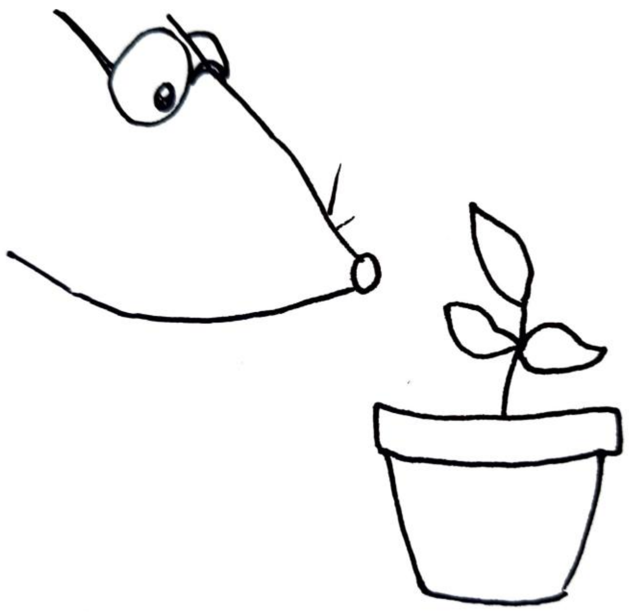
Figure 1: Rats stopped at the red marker, and we measured the stopping distance.

List of Figures

Statistics

Analyzing Plant Growth

```
plant_data <- PlantGrowth
```



Control
Water

Treatment 1
Water + Herbicide

Treatment 2
Water + Fertilizer

0. Explore data with **str()** and **nrow()**

```
plant_data <- PlantGrowth

str(plant_data)
```
```

```
'data.frame':	30 obs. of  2 variables:
 $ weight: num  4.17 5.58 5.18 6.11 4.5 4.61 5.17 4.53 5.33 5.14 ...
 $ group : Factor w/ 3 levels "ctrl","trt1",..: 1 1 1 1 1 1 1 1 1 1 ...
```

```
nrow(plant_data)
```
```

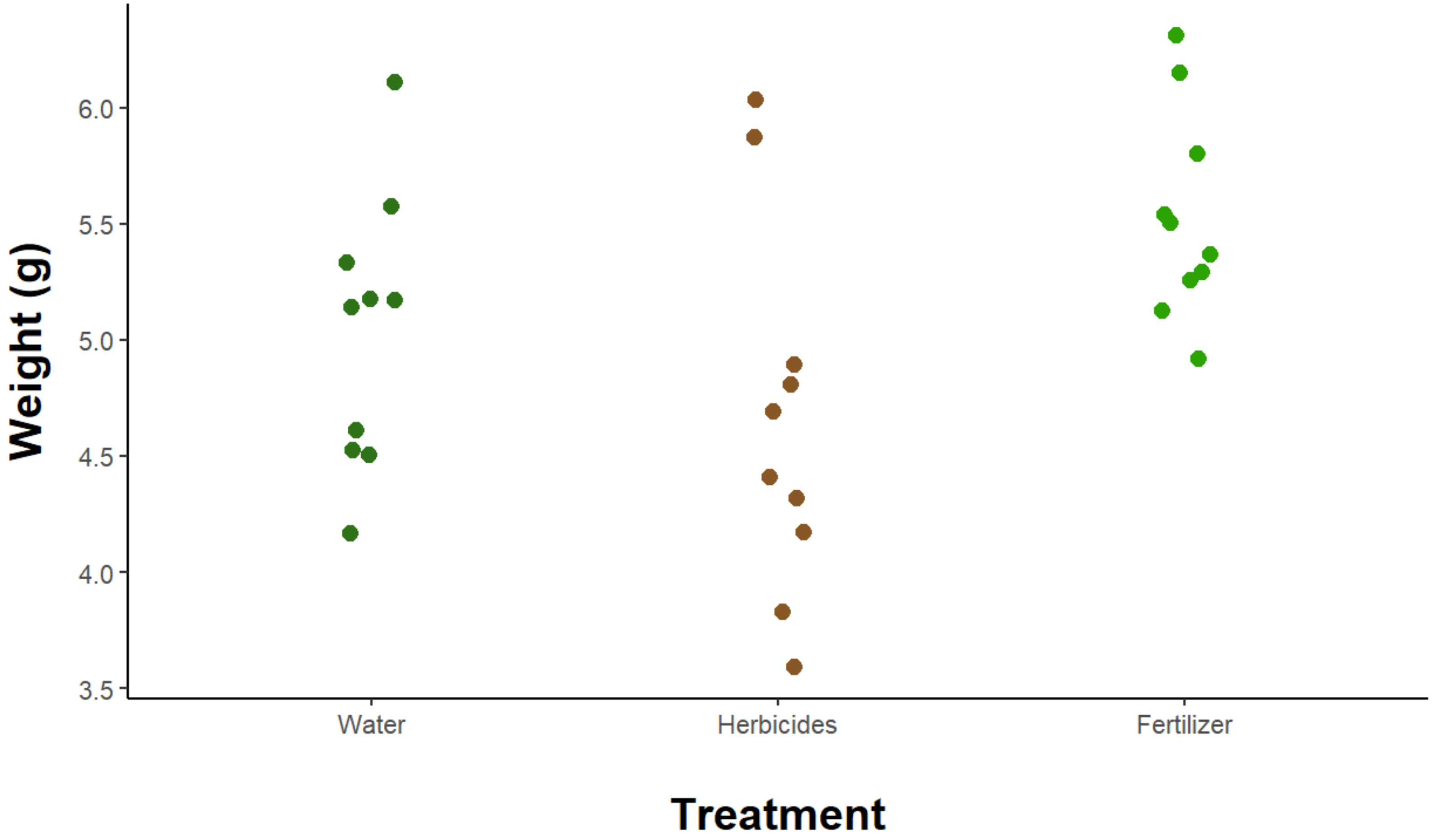
```
[1] 30
```

= 30 plants

Use chunk option **include = FALSE** to hide these printouts in your paper

1. Plot data

```
plant_data %>%
  ggplot(aes(x = group, y = weight, color = group)) +
  scale_color_manual(values = c("#2f7318", "#875826", "#2ca303")) +
  scale_x_discrete(labels = c("Water", "Herbicides", "Fertilizer")) +
  labs(x = "Treatment", y = "Weight (g)") +
  guides(color = "none") +
  geom_jitter(width = 0.07, size = 2.2)
```



2. Choose analysis

One-way Anova

Assess effect of treatment on growth

3. Test assumptions

```
plant_data_aov <- aov(weight ~ group, data = plant_data)
shapiro.test(resid(plant_data_aov))

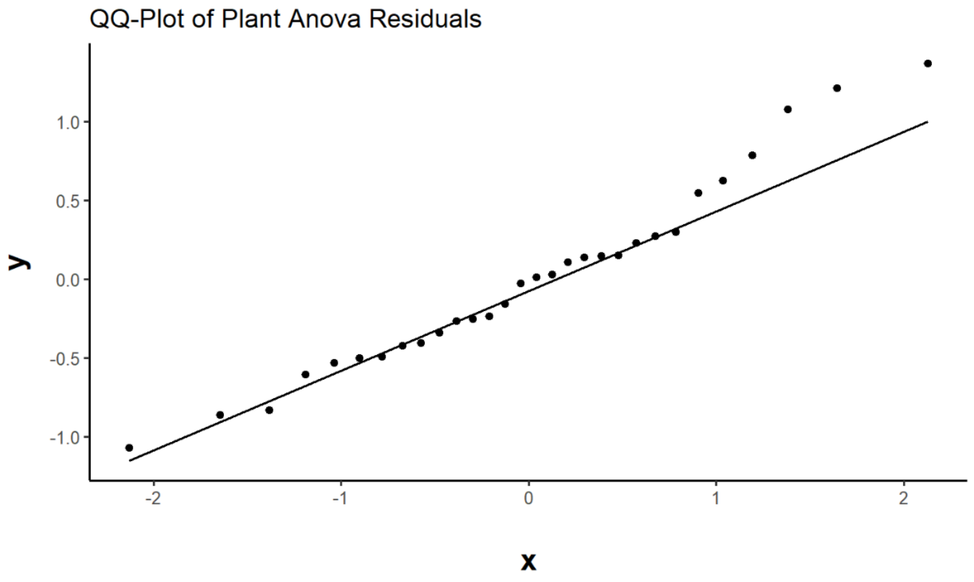
ggplot(plant_data_aov, aes(sample = resid(plant_data_aov))) +
    stat_qq() +
    stat_qq_line() +
    labs(title = "QQ-Plot of Plant Anova Residuals")
```

Normality of residuals

```
        Shapiro-Wilk normality test

data:  resid(plant_data_aov)
W = 0.96607, p-value = 0.4379
```

Data are normal!



4. Interpret results

```
plant_aov_summary <- summary(plant_data_aov)
plant_aov_summary
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
group        2  3.766  1.8832   4.846 0.0159 *
Residuals   27 10.492  0.3886
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Treatment affects plant growth!

5. Report results in text

```
nrow(plant_data)
```
```

```
[1] 30
```

Backticks (above tab key) make a “mini chunk”

```
We weighed `r nrow(plant_data)` plants.
```

We weighed 30 plants.
```

6. Extract p-values

```
View(plant_aov_summary)
```

Name	Type	Value
plant_aov_summary	list [1] (S3: summary.aov, listof)	List of length 1
[[1]]	list [2 x 5] (S3: anova, data.frame)	A data.frame with 2 rows and 5 columns
Df	double [2]	2 27
Sum Sq	double [2]	3.77 10.49
Mean Sq	double [2]	1.883 0.389
F value	double [2]	4.85 NA
Pr(>F)	double [2]	0.0159 NA

Click on green arrow next to p-value

```
> View(plant_aov_summary)
> plant_aov_summary[[1]][["Pr(>F)"]]
```

6. Extract p-values

```
> View(plant_aov_summary)
> plant_aov_summary[[1]][["Pr(>F)"]]
[1] 0.01590996         NA

> plant_aov_summary[[1]][["Pr(>F)"]][[1]]
[1] 0.01590996
```

Use **[[1]]** to select the first p-value, which is associated with *treatment*

```
> round(plant_aov_summary[[1]][["Pr(>F)"]][[1]], 3)
[1] 0.016
```

Round to **3** decimal places

```
Plant growth is affected by chemical exposure
(p = `r round(plant_aov_summary[[1]][["Pr(>F)"]][[1]], 3)`).
```

Plant growth is affected by chemical exposure (p = 0.016)

7. Make publication-quality ANOVA table

```
            Df Sum Sq Mean Sq F value Pr(>F)  
group        2  3.766  1.8832   4.846 0.0159 *
Residuals   27 10.492  0.3886                 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Table 1: Plant growth varies significantly with treatments. ANOVA summary table for a model examining the effect of treatment on plant mass.

Term	df	Sum of Squares	Mean Squares	F-statistic	p-value
Group	2	3.77	1.88	4.85	0.016
Residuals	27	10.49	0.39	NA	NA

7a. Make dataframe of ANOVA results

```
plant_data_aov %>%
  tidy()
```
```

A tibble: 2 x 6

| term <chr> | df <dbl> | sumsq <dbl> | meansq <dbl> | statistic <dbl> | p.value <dbl> |
|---|---|---|---|---|---|
| group | 2 | 3.76634 | 1.8831700 | 4.846088 | 0.01590996 |
| Residuals | 27 | 10.49209 | 0.3885959 | NA | NA |

**tidy()** requires library(broom)

tidy() produces a dataframe instead of a printout
```

7b. Round values & rename columns

```
plant_data_aov %>%
  tidy() %>%
  mutate(
    across(sumsq:statistic,round, 2),
    p.value = pvalString(p.value),
    term = str_to_title(term)
  ) %>%
  rename(
    Term = term,
    'Sum of Squares' = sumsq,
    'Mean Squares' = meansq,
    'F-statistic' = statistic,
    'p-value' = p.value
  )
```

Round values

Capitalize names. **str_to_title()** requires library(stringr)

Make pretty column titles

Term <chr>	df <dbl>	Sum of Squares <dbl>	Mean Squares <dbl>	F-statistic <dbl>	p-value <chr>
Group	2	3.77	1.88	4.85	0.016
Residuals	27	10.49	0.39	NA	NA

7c. Use kable()

```
rename(
  Term = term,
  'Sum of Squares' = sumsq,
  'Mean Squares' = meansq,
  'F-statistic' = statistic,
  'p-value' = p.value
) %>%
kable(
  booktabs = TRUE,
  linesep = '',
  caption = "Plant growth varies significantly with treatments.
      ANOVA summary table for a model examining the effect of
      treatment on plant mass."
) %>%
kableExtra::column_spec(1, width = "1.5in")
```

kable() requires library(knitr)

Uses the booktabs style (horizontal lines around header row and bottom last row)

Don't put a space every 5 rows

Make first column wider

7c. Use kable()

Table 1: Plant growth varies significantly with treatments. ANOVA summary table for a model examining the effect of treatment on plant mass.

Term	df	Sum of Squares	Mean Squares	F-statistic	p-value
Group	2	3.77	1.88	4.85	0.016
Residuals	27	10.49	0.39	NA	NA

7c. Use kable()

Table 1: Plant growth varies significantly with treatments. ANOVA summary table for a model examining the effect of treatment on plant mass.

Term	df	Sum of Squares	Mean Squares	F-statistic	p-value
Group	2	3.77	1.88	4.85	0.016
Residuals	27	10.49	0.39	NA	NA

Customization

1. Customize PDF

Spacing?

Fonts?

Page margins?

Thesis

Contents

List of Figures

Introduction

Text here. . .

Methods

Animals

Reagents

Results

We found that faster cars took longer distances to stop (Figure 2).

2. Improve spacing

```yaml
---
title: "Thesis"
output: pdf_document
fontsize: 12pt


---
```

Photo by Anthony on Pexels

YAML

Yet Another
Markup Language

Thesis

Contents

List of Figures

Introduction

Text here. . .

Methods

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Results

We found that faster cars took longer distances to stop (Figure 2).

2. Improve spacing

```
---
title: "Thesis"
output: pdf_document
fontsize: 12pt
geometry:
- margin = 1in
linestretch: 1.5
---
```

Thesis

Contents

List of Figures

Introduction

Text here. . .

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Results

We found that faster cars took longer distances to stop (Figure 2).

2. Improve spacing

```
---
title: "Thesis"
output: pdf_document
fontsize: 12pt
geometry:
- margin = 1in
linestretch: 1.5
---
```

Thesis

Contents

List of Figures

Introduction

Text here. . .

3. Prepare for printing

Exaggerated binding offset!

```yaml
---
title: "Thesis"
output: pdf_document
fontsize: 12pt
geometry:
- margin = 1in
- bindingoffset = 10mm
linestretch: 1.5
classoption: twoside
indent: true
---
```

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Methods

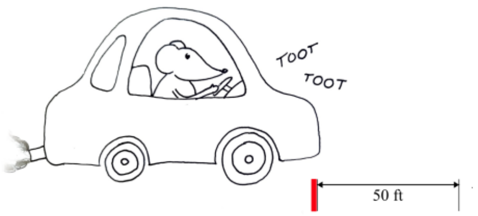

Figure 1: Rats stopped at the red marker, and we measured the stopping distance.

Animals

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2

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Reagents

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Results

Cars Data

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3

LaTeX

(Lay-tech)

4. Modify preamble

Create a new **.tex** file and save it

```
output:
  pdf_document:
    latex_engine: xelatex
    includes:
      in_header: "../Templates/sample-preamble.tex"
```

I used the Templates folder

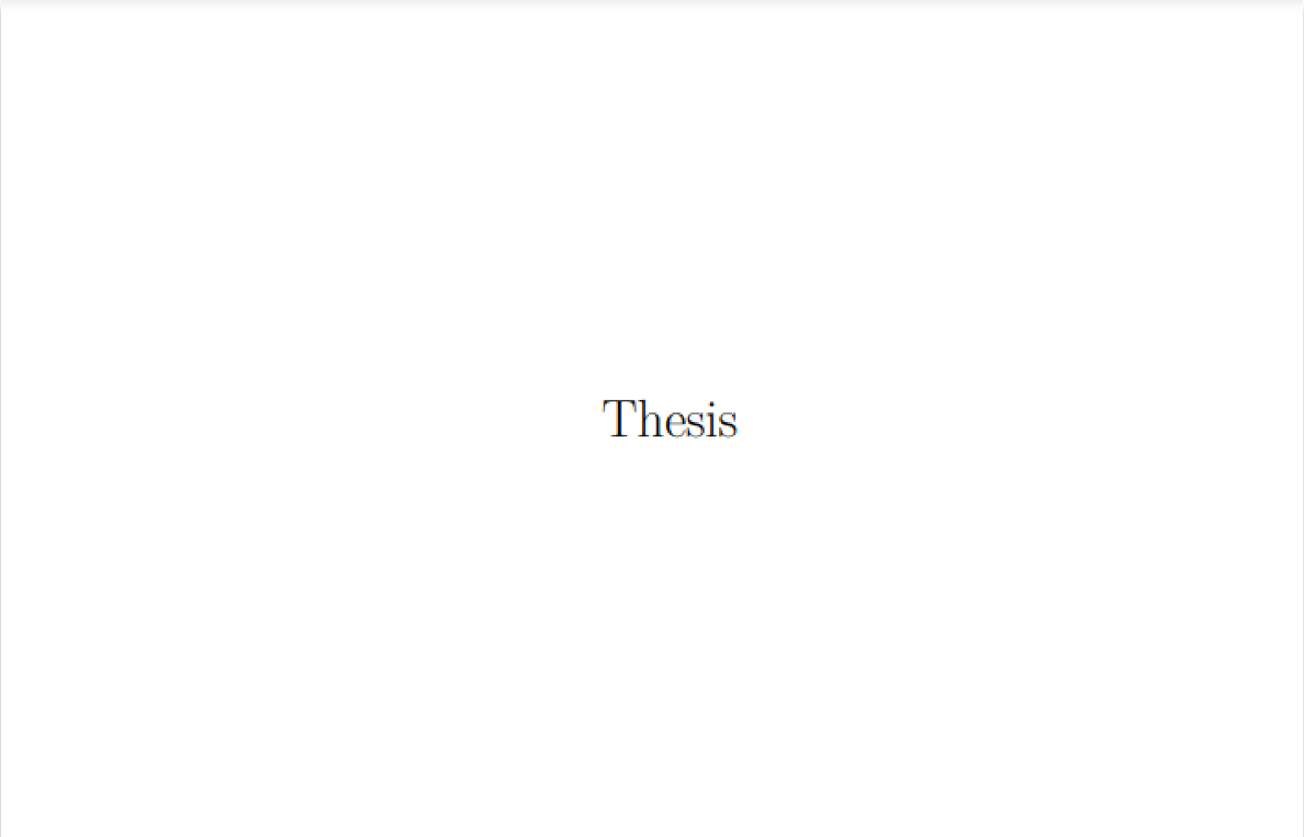
- masters-thesis.Rproj
- README.md
- Thesis
- Figures
- Templates

../ = go up one folder. Reference point is the location of Thesis.Rmd, which is in **Thesis/** folder

4. Modify preamble

```
\pagenumbering{gobble} % Hide page number on title page

\usepackage{titling}
\pretitle{\linespread{1}\begin{center}\vspace{2cm}\Huge}
```



4. Modify preamble

```
\usepackage[skip=20pt plus1pt, indent=0pt]{parskip}

\usepackage[nottoc]{tocbibind}

\usepackage{sectsty}

\newcommand{\BIG}{\fontsize{36}{43}\selectfont}

\makeatletter
\renewcommand\section{\@startsection {section}{1}{\z@}%
        {-7ex \@plus -1ex \@minus -.2ex}%
        {7ex \@plus.2ex}%
        {\newpage\vspace*{-0.2in}\normalfont\BIG\SS@sectfont}}
\renewcommand\subsection{\@startsection{subsection}{2}{\z@}%
        {-3.5ex\@plus -1ex \@minus -.2ex}%
        {3.5ex \@plus .2ex}%
        {\normalfont\LARGE\itshape\SS@subsectfont}}
\makeatother
```

Change paragraph spacing

Include list of figures in TOC

Make a new font size

Increase padding around headers and make headings larger

Make heading 1 start on a new page

Introduction

Heading 2

Lorem ipsum dolor sit amet, consecte
Nullam nec mi et neque pharetra solli
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Introduction

Heading 2

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Fonts

1. Set custom font

```
output:
  pdf_document:
    latex_engine: xelatex
    includes:
      in_header: "../Templates/sample-preamble.tex"
fontsize: 12pt
geometry:
```

Must specify **XeLaTeX** engine in your YAML or it will NOT work!

EB Garamond

abcdefghijklmnopqrstuvwxyz 0123456789

8. Custom font

```
\setmainfont{EBGaramond}[
  Path=../Templates/,
  Extension=.otf,
  UprightFont=*12-Regular,
  ItalicFont=*12-Italic,
  RawFeature=+calt,
]
```

Put .otf or .ttf files in **Templates/**

EBGaramond12-Regular.otf
EBGaramond12-Italic.otf

Québec,

Advanced typography,

e.g. contextual alternates

Introduction

Heading 2

Lorem ipsum dolor sit amet, consectetue
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Heading 3

Introduction

Heading 2

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consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesqu
placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandi
blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia loren
sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

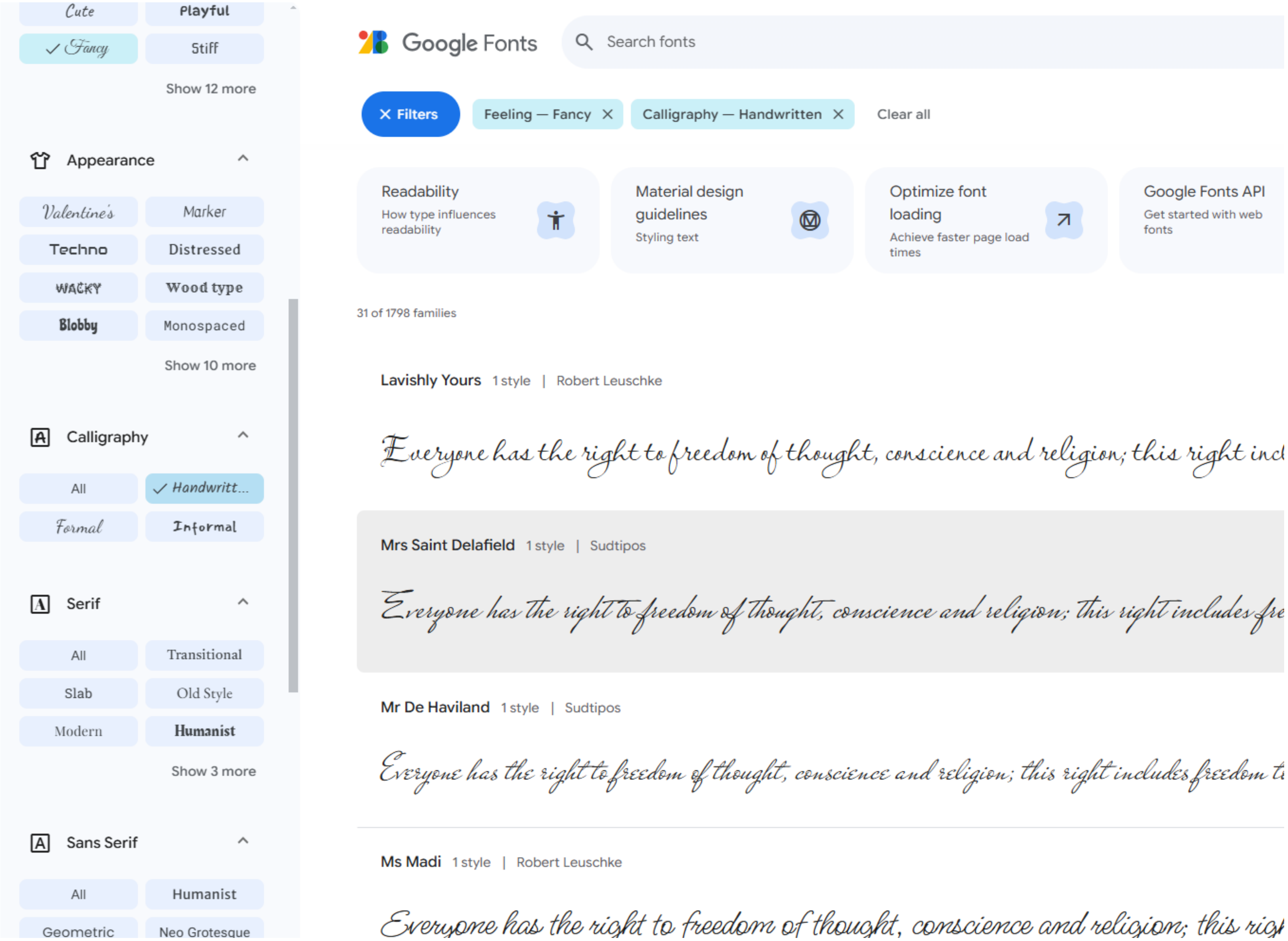
Heading 3

Google Fonts

https://fonts.google.com/

Download a font

Place **.otf** or **.ttf** file in Templates/ folder



Google Fonts

```
\setmainfont{Birthstone}[
  Path=../Templates/,
  Extension=.ttf,
  UprightFont=*-Regular,
  ItalicFont=*-Regular
]
```

Introduction

Heading 2

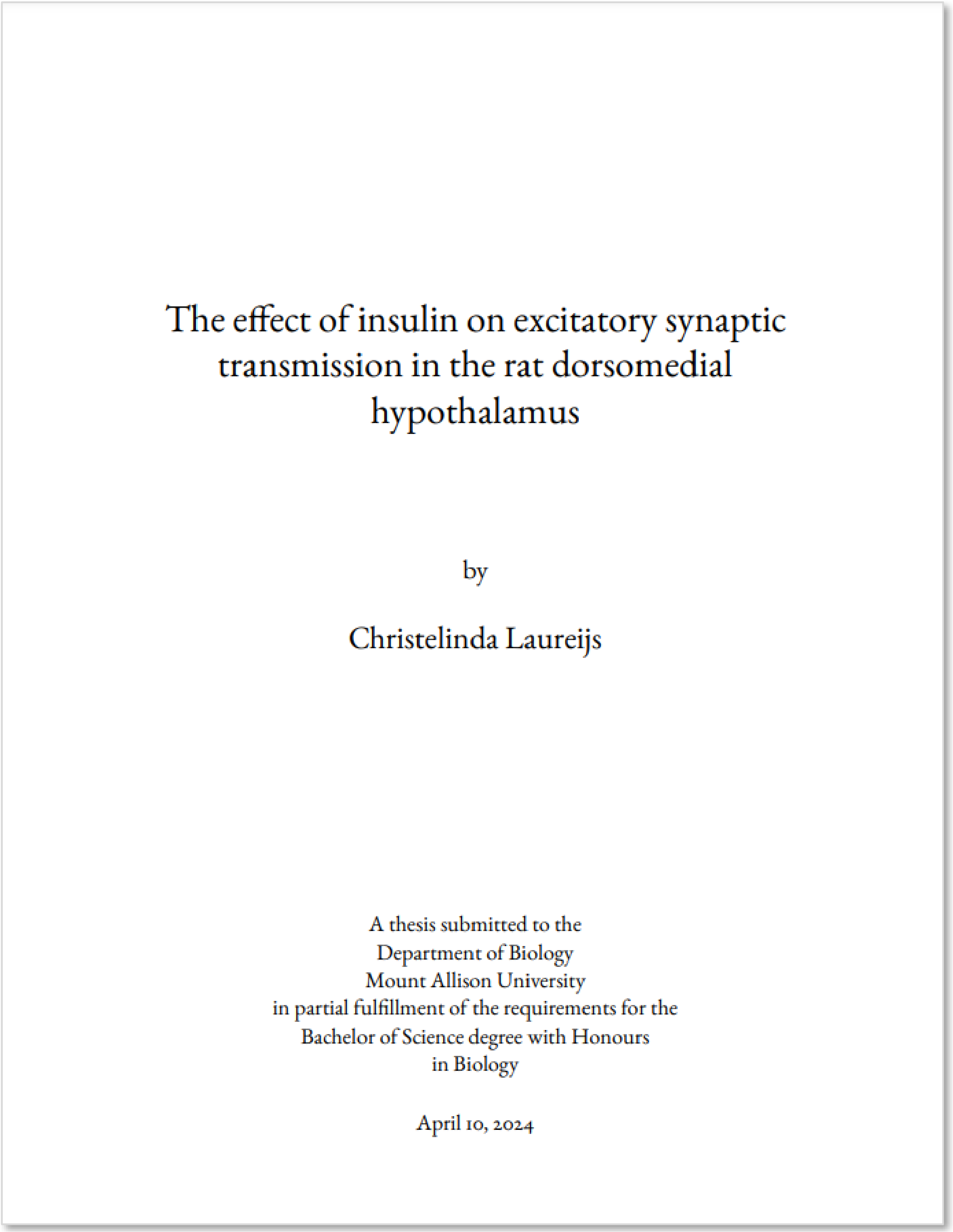
Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Heading 3

9. Make Title Page

```
---
title: The effect of insulin on excitatory synaptic transmission in the rat dorsomedial
  hypothalamus
```

Put title in YAML



The effect of insulin on excitatory synaptic transmission in the rat dorsomedial hypothalamus

by

Christelinda Laureijs

A thesis submitted to the
Department of Biology
Mount Allison University
in partial fulfillment of the requirements for the
Bachelor of Science degree with Honours
in Biology

April 10, 2024

```latex
\begin{centering}
\begin{spacing}{1}
\Large

by
\LARGE

Christelinda Laureijs

\end{spacing}
\end{centering}

\large
\vspace*{\fill}


\begin{centering}
\begin{spacing}{1}
A thesis submitted to the\\
Department of Biology\\
Mount Allison University\\
in partial fulfillment of the requirements for the\\
Bachelor of Science degree with Honours\\
in Biology

April 10, 2024
```

by

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in Biology

April 10, 2024

Decorative symbols with *adforn*

Add this code to the preamble: `\usepackage{adforn}`

`\adforn{19}`

1:
2:
3:
4:
5:

16:
17:
18:
19:
20:

31:
32:
33:
34:
35:

Decorative symbols

```
\adforn{20} *COLOPHON* \adforn{48}

\normalsize

This thesis was typeset in R Markdown and the te
```

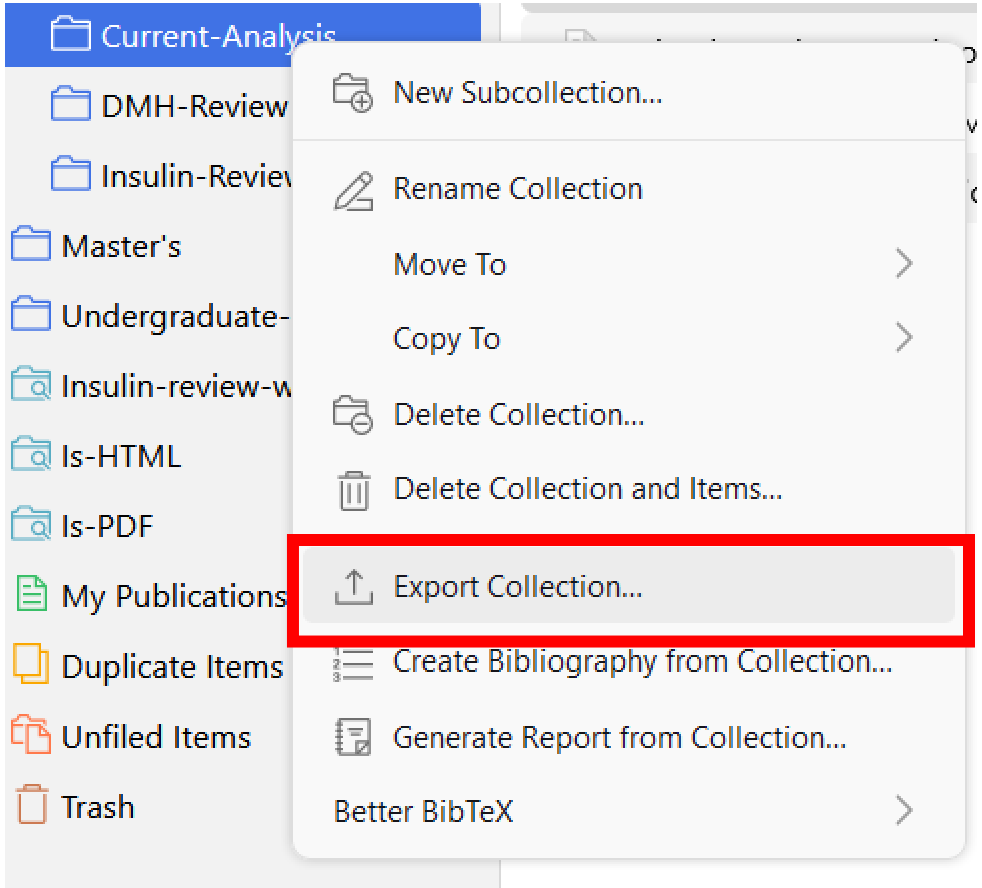
COLOPHON

This thesis was typeset in R Markdown and the text

produce this document are available at *https://githu*

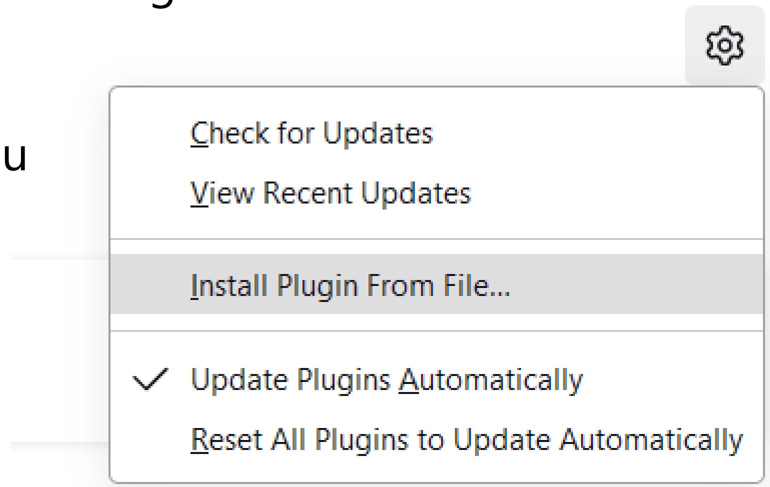
Citations

1. Install BetterBibTeX

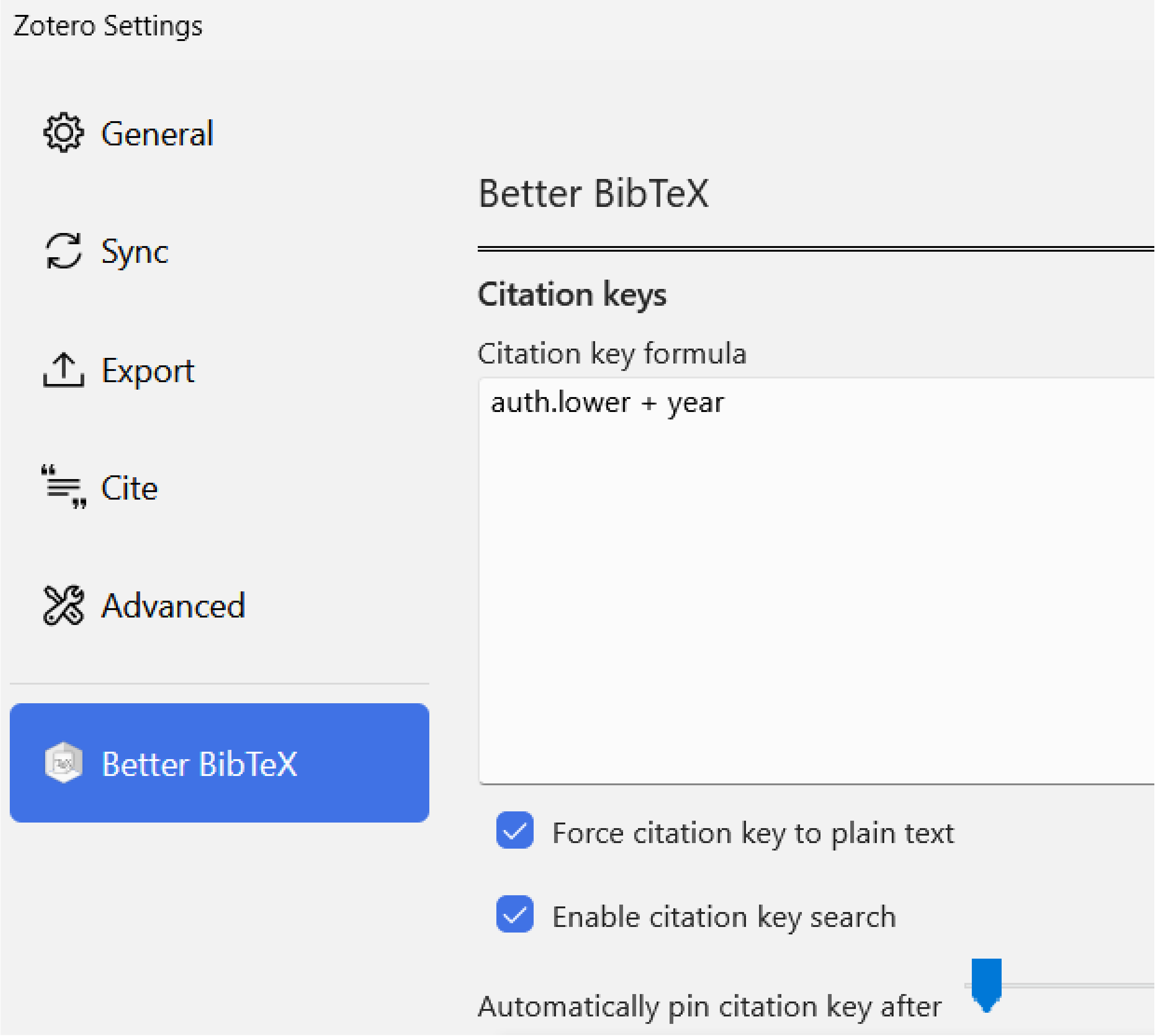


Automatically-updating .bib file in Zotero

- Use the Better BibTeX plugin (https://retorque.re/zotero-better-bibtex/installation/)
- Tools -> Plugins -> Install Plugin from File
- Choose the **.xpi** file you downloaded



2. Set BetterBibTeX citation keys

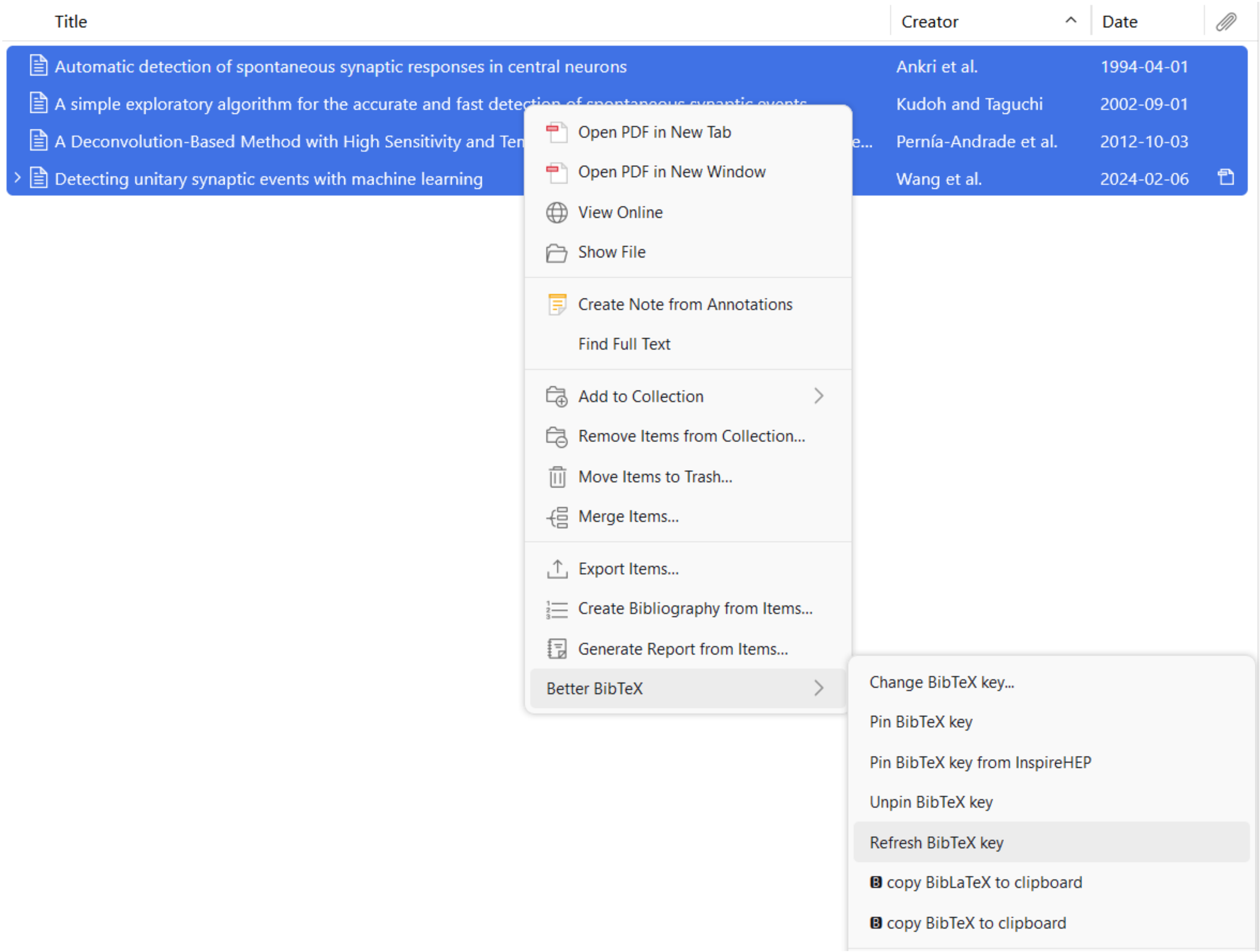


Edit -> Preferences -> Better BibTeX

auth.lower + year

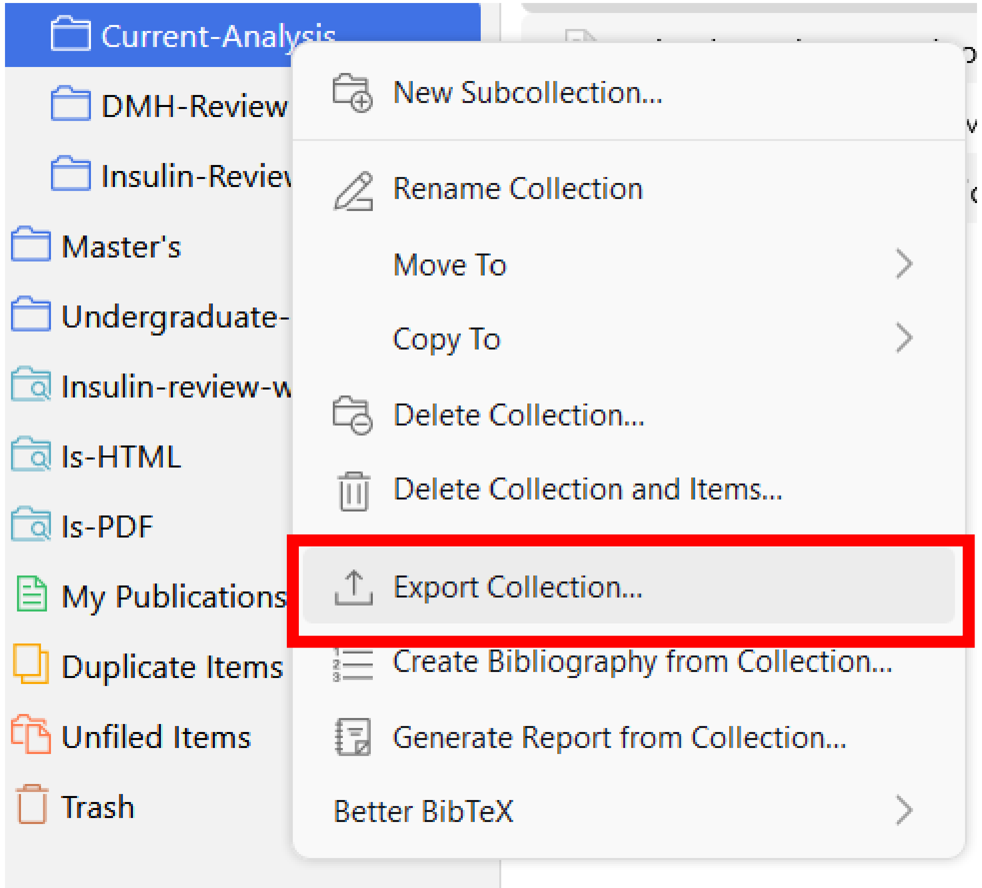
wang2024

3. Refresh all keys in document (only needed after major changes)



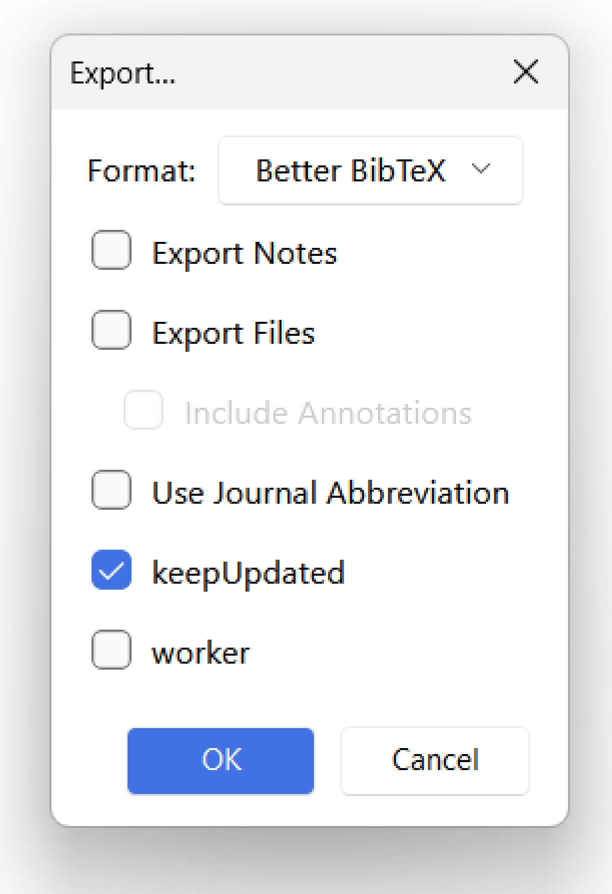
Select all entries, right-click -> Better BibTeX -> Refresh BibTeX key

4. Export Zotero references as .bib



Save in the same folder as the thesis

Check **keepUpdated**



5. Citations in Zotero

```
  pdf_document:
    latex_engine: xelatex
    includes:
      in_header: "../Templates/sample-pre
bibliography: "sample-bib.bib"
fontsize: 12pt
geometry:
  - margin = 1in
```

```
Here is a citation [@ankri1994]
```

Here is a citation (Ankri et al. 1994)

```
Here are multiple citations [@ankri1994; @wang2024]
```

References

Ankri, N., P. Legendre, D. S. Faber, and H. Korn. 1994. "Automatic Detection of Spontaneous Synaptic Responses in Central Neurons." *Journal of Neuroscience Methods* 52 (1): 87–100. https://doi.org/10.1016/0165-0270(94)90060-4.

6. Set location of bibliography

```
# References

::: {#refs}
:::

# Additional Information
```

What you'll look like when you finish your awesome paper!